ZCAS UNIVERSITY

FINAL EXAMINTATION

CAS5831 – DATA MANAGEMENT AND PROCESSING

DATE: 28TH MAY 2025

TIME: 14:00 – 17:00HRS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer Two (2) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A – COMPULSORY

Question ONE

(a) What are the main objectives of data management?

[10 marks]

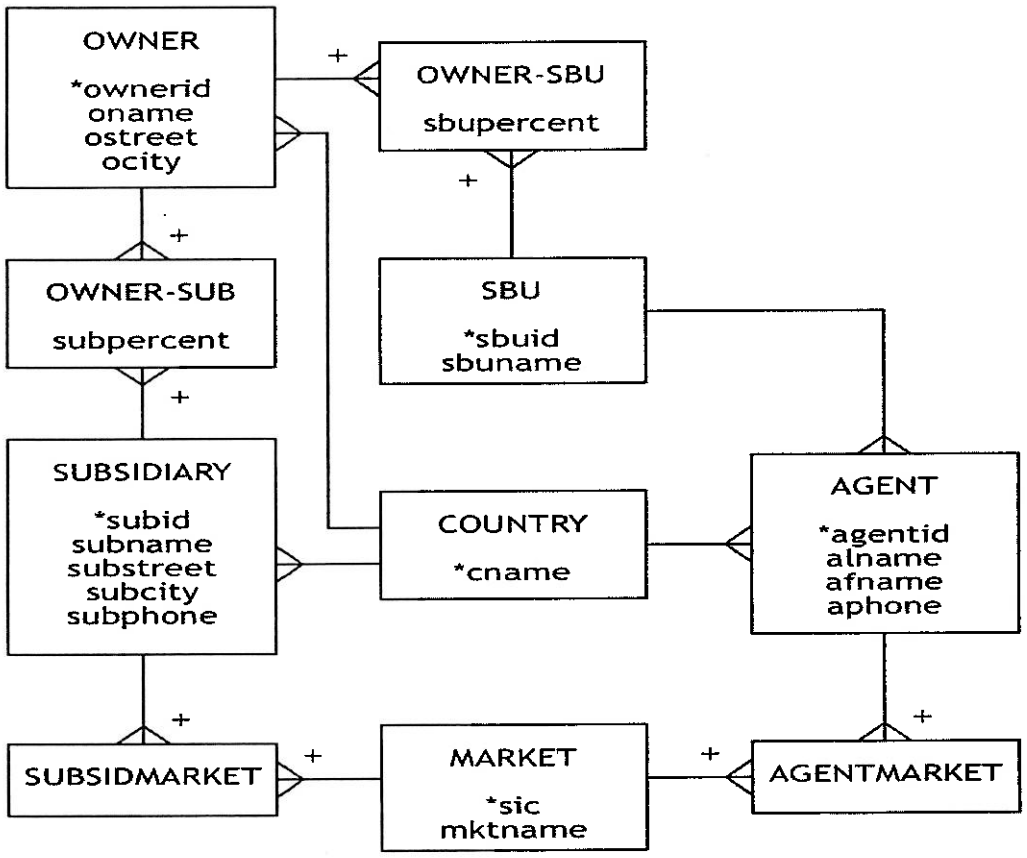


Figure 1: Part of the ER Diagram of the GT

(b) Using the Standard Structured Query Language (SQL), write SQL statements that can be used to implement the GT database model shown in Figure 1. Your statements should show all constraints as table-level constraints for ease data management.

[40 marks]

(Total 50 marks)

SECTION B: Attempt TWO questions from this section

Question TWO

(a) How does effective data management contribute to organizational decision-making?

[10 marks]

(b) What are common challenges faced in managing large datasets?

[15 marks]

(Total 25 marks)

Question THREE

(a) Why is data quality important in data management?

[15 marks]

(b) How does data governance differ from data management?

[10 marks]

(Total 25 marks)

Question FOUR

Discuss the concept of a data model, and explain the different types of data models used in Data Management and Processing?

(Total 25 marks)

END OF CAS5831 FINAL EXAMINATION PAPER JUNE 2025